Rewriting political content to improve LLM judge scores

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Abstract

Large language models (LLMs) increasingly decide which resumes get screened, which websites a chatbot consults, and how documents are ranked. This creates an incentive to optimise content for favourable LLM evaluation while preserving its meaning for human readers — a class of techniques we call rewriting methods. We argue that an effective rewriting method must work when the LLM judge and its prompt are unknown, preserve the document's original meaning, and beat trivial baselines such as asking an LLM to paraphrase; existing studies rarely satisfy more than one of these.

We develop a framework for the rewriting problem and apply it to political opinion paragraphs across four methods (naïve and literature-informed prompting, Best-of-N selection, and GRPO fine-tuning), three small open-weight rewriters and one frontier rewriter, scored by three in-training and six held-out LLM judges on two criteria. Rewriting is partly robust to unseen judges and prompts; gains concentrate on already-low-rated documents and high-quality originals are often degraded; and a prompt-engineered frontier model out-performs every bespoke method, suggesting that improving genuine document quality may be the most robust route to higher LLM-judged scores.

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1 Introduction

Large language models (LLMs) are increasingly used to evaluate, filter and rank documents and websites. They are integrated into systems which screen resumes, re-rank search results [1], and gauge the relevance of documents [2]. Similarly, when a chatbot user requests information about a topic, they rely on an LLM to filter hundreds of websites on their behalf [3]. In each example, an LLM is a gatekeeper between content and audience, providing a strong incentive for creators of resumes, websites, and political material to optimise their content so that it is favourably evaluated by LLMs. This has given rise to startups promising to improve the visibility of their clients' content in LLM responses via 'answer engine optimisation' [4, 5], and prompted research investigating the extent to which LLM evaluators can be gamed [6].

This study focuses on *rewriting methods*. A rewriting method edits a document so that LLM judges score it more favourably, while preserving its meaning. The meaning-preservation requirement distinguishes rewriting methods from prompt-injection attacks, in which the attacker does not care about preserving the meaning of their content [6, 7]. While rewriting methods have been relatively under-researched, they are widely applicable: there are many situations in which one wishes for their document to be favourably assessed by an LLM, but still preserve its meaning for human consumption. For example, a paper passing an automatic LLM evaluation still needs approval from a human peer reviewer. For another example more relevant to our study, if someone wishes to optimise political content for promotion by LLM-powered recommenders, they still want their political message to be preserved.

Research into rewriting methods has been largely unsystematic. We argue that a rewriting method should be assessed against three criteria. Firstly, it should be effective even when the rewriter lacks access to the judge model and prompt. Secondly, it should preserve the original meaning of the document. Finally, it should outperform trivial baseline methods. Existing studies of rewriting methods have not addressed these criteria. For example, both Kaneko [8] and Pan et al. [9] demonstrated that rewriting methods retained *some* effectiveness when tested on a held-out judge model, but did not compare across methods (including trivial rewrites), systematically investigate meaning-preservation, or determine effectiveness when the judge prompt was unknown. Wu et al. [10] provide the closest work to our study. While Wu et al. focused on optimising content for search ranking (rather than LLM judge scores, as we do), they thoroughly treat generalisation and comparison against other methods, and provide a (somewhat limited) treatment for meaning preservation. Notably, our work addresses a question which has not been treated in the literature: whether rewriting methods are uniformly effective, or primarily benefit already low-quality documents.

We develop a general mathematical formulation for the rewriting problem, and use it to develop a practical evaluation framework for the specific example of optimising political writing. Our evaluation spans 4 rewriting methods, 3 rewriter base models, 3 in-training judges, and 6 held-out evaluation judges. Our study focuses on three research questions:

1. **RQ1**: Are rewriting methods effective when evaluated by held-out judges and held-out prompts?
2. **RQ2**: Do rewriting methods uplift all documents, or are gains concentrated on low-quality ones?
3. **RQ3**: When rewriting a document, is there a trade-off between boosting LLM judge scores, and preserving the document's original meaning?

We find that when using small base models for rewriters, rewriting effectiveness is reduced but not eliminated when assessed by held-out judges, and further reduced when assessed using held-out prompts. We also find that documents which are already regarded favourably by LLM judges receive significantly reduced benefits from rewriting, and have a high likelihood of their score being degraded by rewriting. To address this risk, we find that LLMs are sufficiently good predictors of one anothers' ratings, that they can be used to estimate when rewriting is likely to be helpful or harmful. We also find that when model capability is fixed, there is a trade-off between meaning preservation and rewriting effectiveness. Finally, we find that a prompt-engineered frontier model outperforms all bespoke methods we investigated. This last finding suggests that prompt-engineered high-capability models might be the most cost-effective approach to rewriting methods, and also points to the fact that genuinely improving a document's quality might be the most robust method for improving evaluation scores.

2 Background

In a 'rewriting problem', a rewriter aims to improve how a document is scored by an LLM being used as an evaluator, by rewriting the document while preserving its meaning. We define D as the space of all possible documents. A rewriter $R : D \to D$ maps an original document $x \in D$ to a rewritten document $R(x) \in D$.

A semantic similarity function $S : D \times D \to [0, 1]$ measures how similar two documents are in meaning. If $S(x, y) = 1$ for two documents x and y, then they are identical.

The collection of all possible LLM judges is denoted as $\mathcal{J}$. A judge $J \in \mathcal{J}$ maps a document x to its score $J(x) \in \mathbb{R}$.

The aim of a rewriter R is to maximise $J(R(x))$, the judge score of a rewritten document, while maintaining semantic similarity. This optimisation problem may be stated as

$$\text{maximise } J(R(x)) \text{ such that } S(x, R(x)) > \epsilon, \tag{1}$$

where $\epsilon \in [0, 1]$ is a tolerance threshold defining how much semantic drift the attacker is willing to tolerate.

This assumes the judge J is known, which is generally not the case. To account for uncertainty, we define a probability distribution μ over judges, encoding the rewriter developer's beliefs about which judge is being used.

The rewriter then maximises the expected score under this uncertainty:

$$\text{maximise } \mathbb{E}_{\mu}[J(R(x))] \text{ such that } S(x, R(x)) > \epsilon. \tag{2}$$

Note that if the rewriter developer is completely certain that a specific judge will be used, then Equation (2) simplifies to Equation (1) (see Appendix A.1 for proof). For this reason, we refer to Equation (1) as the known-judge formulation, and Equation (2) as the unknown-judge formulation.

Because rewriter developers typically do not know the combination of prompt and LLM that will be used to judge their content, the unknown-judge formulation is almost always the more appropriate formulation to evaluate rewriting methods.

3 Methods

3.1 Experiment overview

We evaluated a variety of rewriting methods against a variety of LLM judges. We used a corpus of human-written responses to political propositions. We rewrote these documents using 3 different rewriting methods. Some 'judge-in-the-loop' rewriting methods used feedback from judges reserved for training ('training judge panel'). Others ('judge-free') did not. We used 3 different LLMs as the base for rewriter models, and 3 different LLMs were in the training judge panel.

Two attributes were set as LLM-judge evaluation targets: clarity and informativeness. Across different combinations of rewriting methods, training judges, and target criteria, we generated $84{,}966$ rewrites.

We evaluate these rewrites in two ways. Firstly, we used a hold-out set of 6 evaluation judges ('evaluation judge panel') to score all rewrites on both criteria. Together with the 3 judges in the training-judge panel, this resulted in $1{,}280{,}908$ scorings. Secondly, we developed two methods for estimating semantic similarity and investigated whether rewritten content deviated from the original in meaning.

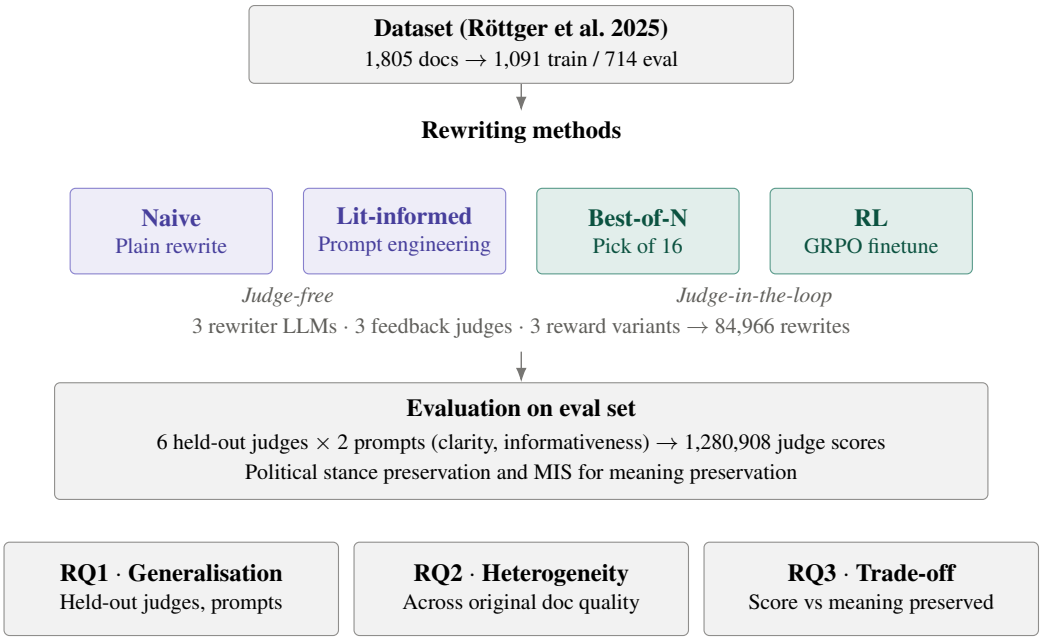

Figure 1: Study design.

3.2 Data

We use a filtered corpus of documents from Röttger et al. [11] for our experiments. The full corpus consists of 10008 documents written in response to 100 UK-related political propositions, such as "The UK should increase taxes on sugary drinks." Half the documents were written by humans selected from a Prolific UK representative sample, and half were written by LLMs. We selected human-written responses to the 40 most controversial topics, yielding 1805 documents. We split these 60/40 into a training dataset ($n = 1091$) and an evaluation dataset ($n = 714$) using a stratified sampling strategy to ensure balance across propositions (see Appendix A.2 for details). All evaluation metrics were calculated using the evaluation dataset.

3.3 Measuring rewriting effectiveness

Evaluating the effectiveness of a rewriter R requires approximating $\mathbb{E}_\mu[J(R(x))]$, which we estimated as the average judge score across an evaluation panel of LLM judges.

Three factors determined which judges to include in our evaluation panel. Firstly, if a judge was used as a reward signal when training R, that they should not be included in the evaluation panel. This was to prevent the evaluation from rewarding a rewriting method which over-fit to the judge(s) used in training. Secondly, we used a diverse range of judges, including 3 judges with similar parameter sizes to those used for feedback, but also including 3 judges with much higher parameter count. Finally, we are constrained by cost and time, and we choose models that could either be run using less than 20GB of VRAM or were cost-effective on OpenRouter. The resulting evaluation panel is summarised in Table 1.

Table 1: Evaluation judge panel.

Judge	Parameters	AAII	Released	Inference
Mistral-7B v0.3 [12]	7 B (dense)	7 [13]	May 2024	local
Phi-3.5-mini [14]	3.8 B (dense)	10 [15]	Aug 2024	local
Command-R 7B [16]	7 B (dense)	—	Dec 2024	local
Qwen3.5-35B-A3B [17]	35 B / 3 B (MoE)	37 [18]	Feb 2026	OpenRouter
Gemini 2.5 Flash-Lite [19]	undisclosed	13 [20]	Jul 2025	OpenRouter
MiMo-V2-Flash [21]	309 B / 15 B (MoE)	30 [22]	Dec 2025	OpenRouter

We used two prompts, corresponding to two different evaluation criteria: clarity and informativeness. See Appendix C.1 for prompts and rationale relating to their design.

To evaluate rewriter effectiveness, we used two metrics: the average change in judge score between the original document and the rewritten document, and the percentage of rewritten documents with a higher average score than the original document. The first metric describes overall effectiveness, while the latter estimates the likelihood of the rewriter being counterproductive.

3.4 Measuring semantic similarity

The choice of semantic similarity function S depends on context. We developed one application-specific method and one general method.

Stance preservation Firstly, we estimated the extent to which rewritten document expressed the same stance towards the proposition that the original document was written in response to. Recall that our dataset consists of political propositions and documents written in response to them. These documents had been annotated with a rating from 0-100 ('agreement_score') indicating the extent to which the document disagreed (0) or agreed (100) with the proposition. We trained a model which takes in an input tuple of $(\text{proposition}, \text{document})$ and returns an estimated agreement_score.

This model is robust: when evaluated on documents responding to propositions entirely excluded from training, it had a mean absolute error (MAE) of 7.1, and the 95th percentile of its absolute error (p95AE) was 19.8. Finally, when it was retrained on only human-written documents and evaluated only on AI-written documents, MAE and p95AE increased only slightly to 8.0 and 22.1.

We used this model to estimate agreement_score for all original and rewritten documents. These estimated agreement_scores were then used to calculate several metrics. Our primary metric is stance-preservation, which is calculated in the following way. A document with agreement_score higher than 75 has had its stance preserved if the rewritten version has agreement_score higher than 50; similarly a document with agreement_score lower than 25 has had its stance preserved if the rewritten document has agreement_score lower than 50. Documents with original stance between 25 and 75 are excluded from the calculation. Stance-preservation rate is the percentage of documents whose stance was preserved. Intuitively, stance-preservation rate answers the following question: for someone with a strong opinion, did their original stance survive the rewrite?

Natural language inference The agreement_score model is interpretable and relevant to our specific problem setting, but not generally applicable. We also used scores from a natural language inference (NLI) model to measure semantic similarity [23]. A NLI model $NLI : D \times D$ takes an ordered tuple of documents (x, y) as input and returns a score $NLI(x, y) \in [0, 1]$ which measures whether the contents of x 'entails' the contents of y. NLI models are effective at preserving the meaning of documents in paragraphing or rewriting tasks [24]. We adopt the Mutual Implication Score (MIS) from [24] as a measure of semantic similarity which is a general alternative to the agreement_score. It averages forward and backward entailment between the original and re-written documents,

$$\text{MIS}(x, R(x)) = \frac{1}{2}(NLI(x, R(x)) + NLI(R(x), x)). \tag{3}$$

We scale these scores by 100 so that they sit in the range $[0, 100]$.

We used the ModernBERT-large-zeroshot-v2.0 NLI model (Hugging Face repository) to calculate MIS. This model almost achieves SOTA accuracy on NLI benchmarks, meets our context length requirements, and is faster than alternatives [25]. We validated MIS against stance preservation, and found that a MIS score of 50 corresponds to a stance preservation rate of >95% Appendix A.4.

3.5 Rewriting methods

We investigate three categories of rewriting methods. They span two axes: whether the method requires feedback from a judge during use (*judge-in-the-loop* vs *judge-free*), and whether it requires finetuning the rewriter (*finetuning* vs *prompt-only*).

Prompt-only and judge-free We tested two prompts. The first, 'naive', prompts the rewriter to rewrite the paragraph while preserving stance and approximate word count. The second, 'literature-informed', instructs the judge to rewrite the paragraph while applying writing principles that have been found to improve LLM assessments of document quality in previous studies [5, 26, 27]. Prompts are available in Appendix C.2. Both methods used temperature $= 0$.

Prompt-only and judge-in-the-loop We implemented a Best-of-N ('BoN') approach. Multiple rewrites were generated using the literature-informed prompt and temperature $= 1$, and then a single rewrite was selected based on judge feedback and other factors (detailed below). We found diminishing returns for $N > 16$ (Appendix A.5), and thus limited N to 16.

Reinforcement learning We used reinforcement learning ('RL') to fine-tune a base rewriter model. We used the Group Relative Policy Optimisation (GRPO) algorithm, and the Transformers Reinforcement Learning (TRL) library implementation [28]. Training hyperparameters are detailed in Appendix A.6. We mostly used TRL defaults, but vary a few where we found better performance. Along with the feedback function variants detailed below, we used a length penalty, which introduces a quadratic penalty for deviations between the length of the rewritten document and the original document, with an additional cubic penalty if the rewritten length is more than 15% larger than the original length.

Variants in reward function A reward function is required for both the BoN and RL rewriter methods. The simplest method is the score from a single judge. We varied this along two dimensions: using the average score from two judges ('ensemble'), and using an equally-weighted average of judge and NLI scores ('judge+NLI'). These were intended to address separate potential issues: using an ensemble of judges for feedback during training potentially reduces overfitting to a single judge, while incorporating semantic similarity reduces the risk of reward hacking, in which rewritten content is highly scored but completely different in meaning to the original content (or unintelligible). These two variants to the reward function yield 3 variations for each judge-in-the-loop model, all of which we evaluated.

Choice of LLMs for rewriters and feedback judges We evaluated rewriting methods across combinations of 3 base rewriter models and 3 feedback judge models. All experiments used a single-A100 training environment. The base rewriters are listed in Table 2 and the feedback judges in Table 3.

Table 2: Base rewriter models.

Rewriter	Parameters	AAII	Released
Qwen2.5-1.5B-Instruct [29]	1.5 B (dense)	—	Sep 2024
LFM2.5-1.2B-Instruct [30]	1.2 B (dense)	8 [31]	Jan 2026
Gemma-3-1B-it [32]	1 B (dense)	6 [33]	Mar 2025

Table 3: Feedback judge panel, used in BoN candidate selection and as the reward signal during GRPO training.

Judge	Parameters	AAII	Released
Qwen-3.5-9B [34]	9 B (dense)	32[1] [35]	Feb 2026
Llama-3.1-8B-Instruct [36]	8 B (dense)	12 [37]	Jul 2024
Gemma-2-9B-it [38]	9 B (dense)	—	Jun 2024

For judge-in-the-loop rewriting methods (BoN and RL), we tested all combinations of rewriters and feedback models. For example, for the single-judge variant of BoN with Qwen2.5-1.5B-Instruct as the base model, it was trained separately with each of Qwen-3.5-9B, Llama-3.1-8B-Instruct, and Gemma-2-9B-it.

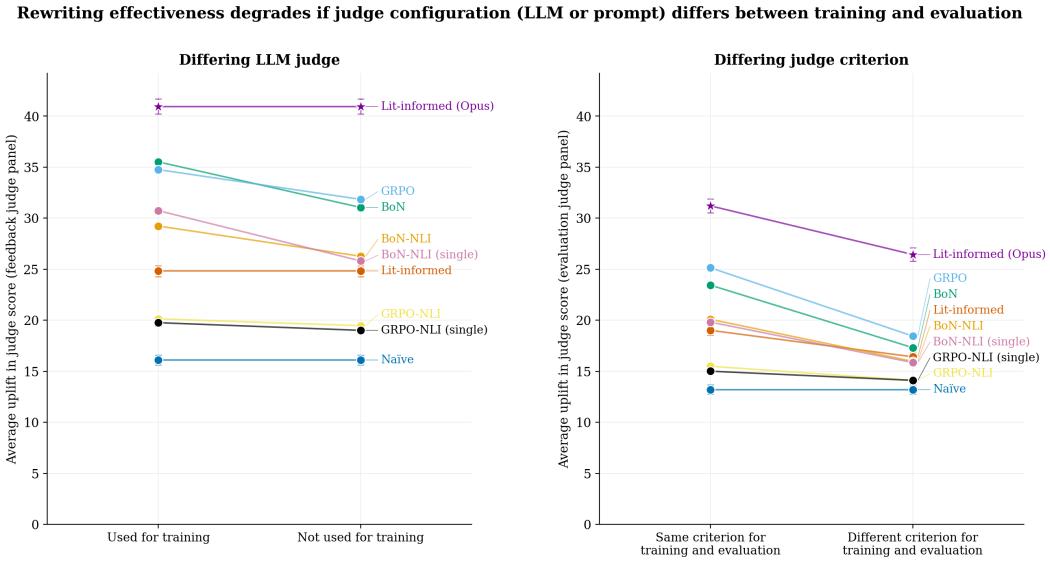

Figure 2: Rewriting effectiveness on different evaluation judge configurations. **Left:** feedback-panel judges (used for training) vs. held-out feedback-panel judges. **Right:** evaluation panel judges scoring the matched (training-target) criterion vs. scoring a different criterion. Y-axes show the mean change in raw judge score (rewrite minus original); error bars are 95% cluster-robust confidence intervals on `rewrite_id`.

4 Results

4.1 Rewriting effectiveness on held-out judges and held-out prompts

Figure 2 measures two different kinds of generalisation error for rewriting methods. The left panel measures degradation in effectiveness due to the judge LLM varying from the feedback LLM. The right panel compares effectiveness on out-of-sample LLM judges scoring the matched criterion against the same judges scoring the criterion that was *not* the rewriter's training target.

Degradation due to differing LLM judge model This was estimated using scores from the feedback panel. Because each judge-in-the-loop rewriter is trained with the feedback from 1 or 2 judges from the feedback panel, but is still scored by all judges in the feedback panel, we have a dataset where every rewritten paragraph is scored by models used for training ('Used for training'), and held-out models ('Not used for training'). Moreover, the exhaustive evaluation across all combinations of feedback judge(s) means that this dataset is balanced, and the difference in average scores between these two categories of scores yields an unbiased estimate of the degradation in effectiveness caused by evaluating using a different LLM than that used for training.

We see degradation among all judge-in-the-loop models. However, the rewriters using an ensemble of 2 judges for feedback have only slightly better performance when evaluated on the held-out judges than variants using only a single judge (denoted by 'single' in the chart). We investigate this further in Appendix B.2, finding that adding a third judge to the ensemble barely increases performance.

Degradation due to differing LLM judge prompt This was estimated using scores from the evaluation panel. All rewrites were scored by all model $\times$ prompt combinations. Hence, rewrites targeting the clarity criterion were still judged by models using the informativeness criterion, and vice versa. This allows us to estimate the reduction in effectiveness caused by a mismatch between the prompt used for training and evaluation. We find that all attack methods reduce in effectiveness, except for Naive, which does not target any particular criterion.

4.2 Relationship between original document quality and rewriting effectiveness

Figure 3 shows how rewriting effectiveness varies with the quality of the original document, where quality is summarised by the mean score of the held-out evaluation judge panel on the original. Both panels share the same x-axis. The left panel reports the mean uplift (rewrite minus original); the right panel reports the share of (rewrite, judge) pairs for which the rewrite scored higher than the original.

Notably, for documents with already-high scores (>80), rewriting often (>50% of the time) degrades the score. Someone deciding whether to use a rewriter would likely want to know whether their content is likely to be receive a better or worse score after rewriting, but under our assumption that they do not have access to the evaluator LLM, they cannot determine if their document would already receive a high score.

We find a possible method to address this issue. It is possible to score a document using one or more LLMs, and relatively robustly predict the likelihood of improvement (see Appendix B.3). This is because LLM judge scores are quite correlated (Appendix B.1), and so a panel of LLMs can roughly estimated the score of another LLM. If the estimated score is high, it suggests that using a rewriter might be counterproductive.

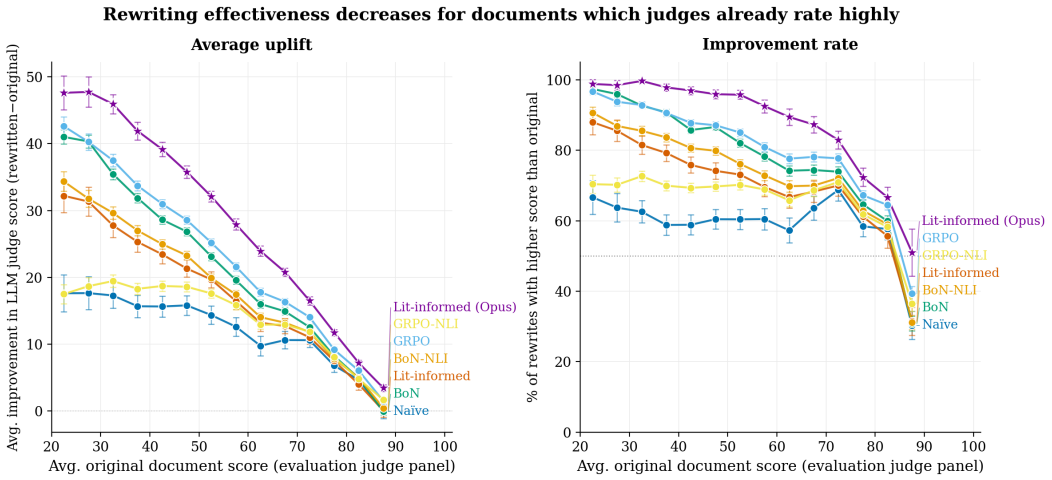

Figure 3: Rewriting effectiveness as a function of original-document quality, measured by the mean of the held-out LLM evaluation judge panel. **Left:** mean change in judge score per rewrite, averaged across the panel. **Right:** share of (rewrite, judge) pairs in which the rewrite is rated above the original. Five-point buckets on the x-axis; error bars are 95% cluster-robust confidence intervals on `rewrite_id`. Across both metrics, every method's effectiveness diminishes as the original document scores higher; the same pattern under the in-panel-LLM and human-labelled scores is reported in Appendix B.3.

4.3 Trade-off between semantic fidelity and rewriting effectiveness

Figure 4 plots each rewriting method on the joint (stance preservation, judge-score uplift) plane, aggregating across all out-of-sample judges and all rewriter base models. The dashed line traces the empirical Pareto frontier across the non-Opus methods. Appendix B.4 reproduces the same chart with each (method, rewriter) cell shown separately. There is a clear tradeoff between preserving the stance of the original document and boosting its score.

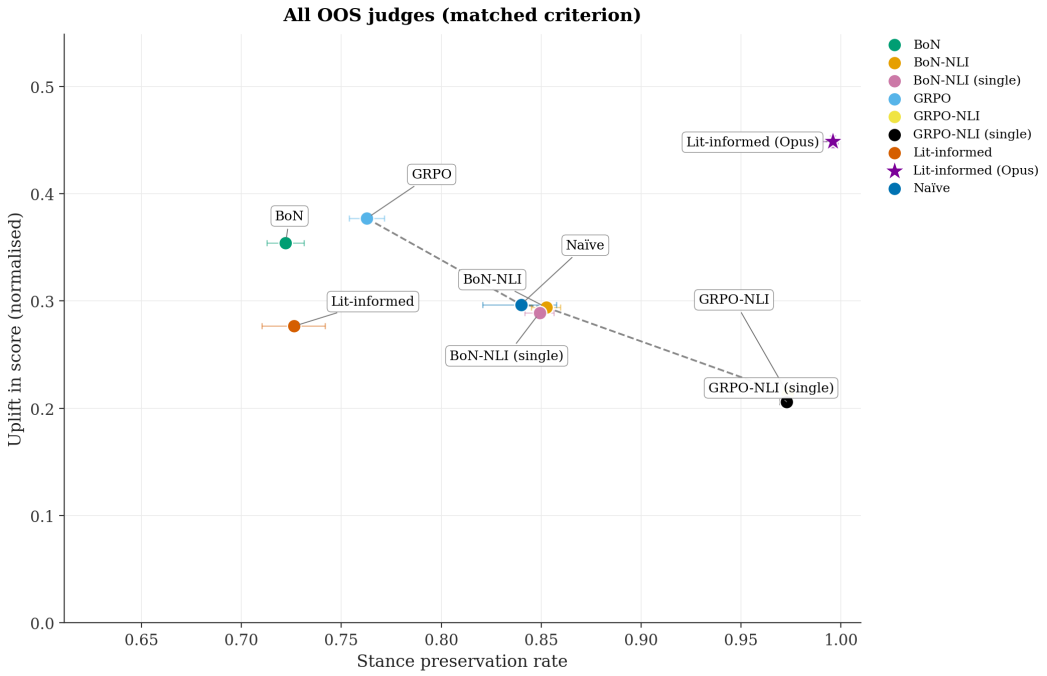

Figure 4: Trade-off between stance preservation and judge-score uplift, aggregated across all out-of-sample judges (matched criterion) and across rewriter base models. The dashed grey line is the Pareto frontier on non-Opus methods (i.e. the maximum achievable uplift at each preservation level). Error bars are 95% confidence intervals on each axis.

5 Discussion

Our main findings are as follows. All methods degraded in effectiveness when evaluated in contexts differing from their training. Improvements from rewriting are concentrated in documents which were already low-scoring, and rewriting an already-highly-scored documents risks degrading its score. However, non-evaluator LLMs can be used to estimate whether rewriting is likely to be beneficial. There is also a clear tradeoff between meaning preservation and rewriting effectiveness. Finally, Opus 4.7 outperforms all models and rewriting methods in terms of preserving meaning and boosting scores.

It is not surprising that rewritten content degraded in score when evaluated using different models and prompts than that used to optimise them. However, it is important. It is difficult to think of circumstances in which a rewriter has access to the judge model. For a rewriting method to be impactful in any way (whether positive or negative), it must be effective in such uncertain circumstances. One obvious way for a prospective rewriter to circumvent this obstacle is to gain information about the judge or prompt used. The implications for designers of LLM evaluation systems is that revealing information about the prompt or LLM judge opens them up to more effective exploitation.

Our findings raise questions about the extent to which rewriting methods are effective simply because they either produce LLM-written content, or produce high-quality content. It is surprising that naive rewriting resulted in a boost of ~13-16 points on average for rewrites that were often barely coherent (see Appendix C.3 for examples). There is substantial evidence that when LLMs are used as judges, they prefer LLM-generated content, regardless of its quality [39–41]. It is likely that at least some score improvements are driven by this factor.

Similarly, the high performance of Opus 4.7 as a rewriter might be due to genuinely increasing document quality. We deliberately used the term 'rewriting method' throughout this paper, rather than 'rewriting *attack*'. This is because it is possible for a rewriter to boost the score of a document without compromising the integrity of the LLM's judgements, by only increasing the quality of its judgements (as long as the judge is an accurate evaluator). This draws an interesting parallel to search engine optimisation. Google has long insisted that the most robust method for promoting websites

in search results is to improve their quality [42]. At the same time, in certain query categories such as product reviews, spammers have exploited weaknesses in Google's search algorithms [43]. The effectiveness of rewriting methods, and the extent to which they are harmful, likely depends on the relative cost-effectiveness of improving quality versus exploiting LLM judges for different content types.

Our focus on generalisation is particularly relevant to this discussion. It might be the case that a single LLM is exploitable, but it is much more difficult to exploit an unknown judge. A generally-effective rewriting method likely employs some mixture of quality improvements and 'leaning into' LLM self-preference bias.[2] It would be valuable to investigate whether more capable and optimised rewriter models produce especially low-perplexity output, given that it has been found that this is potentially the primary driver of LLM self-preference bias [39].

Our findings regarding semantic preservation are particularly striking. In the political context, it is completely unacceptable to a user for their content to flip stance entirely. This suggests that among the non-Opus methods, only the GRPO-NLI method, with its stance preservation rate of 97%, might be suitable for real-life deployment. This validates the use of NLI or similar methods to Other studies of rewriting methods would be well-placed to investigate

Finally, we make a novel contribution by considering how rewriting effectiveness varies across documents. While it is natural that a document with a high original score is less likely to be improved, it is notable that once the original score is sufficiently high, there is worse than a coin-flip chance of worsening the document's score by rewriting it. This is a

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A Methods and Experimental Details

A.1 The unknown-judge formulation is a generalisation of the known-judge formulation

Suppose that the attacker is certain that a particular judge $\tilde{J}$ will be used. Under the unknown-judge formulation, their beliefs about what judges might be used to assess their documents are encoded in the probability distribution μ over judges $\mathcal{J}$. If they are certain that $\tilde{J}$ will be used, then μ is given by

$$\mu(J) = \begin{cases} 1 & \text{if } J = \tilde{J} \\ 0 & \text{otherwise.} \end{cases} \tag{4}$$

Then,

$$\mathbb{E}_{\mu}[J(R(x))] = \sum_{J \in \mathcal{J}} \mu(J) \cdot J(R(x)) = \tilde{J}(R(x)). \tag{5}$$

Plugging this into Equation (2) yields Equation (1). Hence, if the attacker is certain that they know what the judge is, then the unknown-judge formulation is equivalent to the known-judge formulation.

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A.2 Details on document corpus and evaluation & training datasets

Original dataset We use documents from a corpus developed by Röttger et al. [11] for an upcoming publication.

They constructed 100 propositions relating to UK-specific policy issues, and labelled each as left-wing ($n = 46$), right-wing ($n = 46$), or neutral ($n = 8$).

In the first stage of their experiment, they recruited a group of writers ($n = 1500$) from a Prolific UK representative sample. Each writer wrote a response to three propositions, yielding 4500 responses.

Each human-written response was used as the basis for generating a LLM-generated response. This was achieved in the following way. Each human-written response was randomly allocated to one of GPT-4o, Claude Sonnet 4, or DeepSeek V3. Then, with equal probability, one of the following three prompts were used to produce a LLM-generated response:

- Bullets-based ($n = 1404$): all humans wrote bullet points explaining their opinion prior to writing their full response. The LLM was provided with these bullet points and prompted to write a full response.
- Stance-based ($n = 1358$): all humans provided a self-rated level of agreement with the proposition. The LLM was provided only with this agreement score, and prompted to write a response in alignment with this level of agreement.
- Improve ($n = 703$)/rewrite ($n = 677$): The LLM was prompted to improve or rewrite the human-written response.

More details available at [11, p.7].

In the second stage of the experiment, they used a separate pool of participants ($n = 10017$) to rate the perceived level of agreement each paragraph had with the proposition it was written in response to. These participants also rated documents according to criteria such as clarity and informativeness. Raters were not told that some responses were LLM-generated. Each rating was on a scale of 0-100, and each response received 9-19 (average of 10) ratings. We aggregate these human annotations for each document as document-level attribute scores.

The primary attribute used in the current study is agreement_score, which measures the extent to which a document agrees with the proposition it was written in response to. We use intraclass correlation to measure inter-rater reliability. While the reliability of individual rating scores was only $ICC(1) = 0.671$, the reliability of ratings aggregated for each response was much better, with $ICC(10) = 0.953$. This provides support for the validity of using the aggregated human-annotated agreement_score as a ground-truth measure of how much each document agreed with its proposition.

Construction of training and evaluation samples We measured the controversiality of a proposition by using the variance of the human-annotated agreement_score of each proposition. Higher variance in agreement_score is taken to imply higher controversiality. We did this to ensure that

responses spanned a wide range of levels of agreement, rather than all agreeing or disagreeing with propositions. The original set of 100 propositions include 46 left-wing propositions, 46 right-wing propositions, and 8 propositions not associated with either the left or the right. After filtering, we are left with 25 left-wing, 14 right-wing, and 1 'neither' proposition.

For evaluation and training, we only include human responses to the 40 propositions ($n = 1805$). We split these responses 60/40 between training and evaluation. This split was achieved using stratified random sampling. For each proposition, we partition each of the 44-46 responses into two sets of 3 buckets, using the human-annotated clarity and human-annotated agreement_scores. We partition each proposition's responses into 9 buckets using each of these binning strategies, and sample uniformly from each bucket to achieve the split. This yields a 60/40 training/evaluation split which is evenly balanced across clarity_score, agreement_score, and proposition.

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A.3 Agreement-score regressor: training and evaluation

We use a learned regressor to predict the rater-aggregated stance label $y \in [0, 100]$ (where $y = 0$ encodes "argues against the proposition", $y = 100$ encodes "argues for it") for each (proposition, paragraph) pair. The regressor lets us score every rewrite in the eval set without running a fresh rater study, and is the backbone of the agreement-score change metric reported in the main results.

Data. The training data comes from the `paul_data` aggregated rater dataset: 10,008 (proposition, paragraph) pairs across 100 propositions, with a continuous stance label y obtained by averaging `writer_stance` across human raters and rescaling to $[0, 100]$. Three paragraph sources contribute roughly equal volumes: 4,503 human-written paragraphs (`paragraph_type=writer`), 4,503 LLM-written paragraphs across three model authors (`anthropic/claude-sonnet-4`, `deepseek/deepseek-chat-v3-0324`, `openai/chatgpt-4o-latest`, `paragraph_type=model`), and 1,002 human-edited LLM paragraphs (`paragraph_type=edited`). 40 of the 100 propositions are the *controversial* subset used in the main experiments; the remaining 60 are *non-controversial* pre-tests that did not make the controversial cut.

Architecture. `microsoft/deberta-v3-base` (140M parameters) with a single-output regression head, applied to the (proposition, paragraph) sentence pair via the standard sequence-classification API. Inputs are tokenised at `max_length=256`; longer paragraphs are right-truncated. Outputs are clipped to $[0, 100]$ at inference time.

Training procedure. 4 epochs of AdamW with linear warmup over the first 6% of steps, learning rate 1×10^{-5}, weight decay 0.01, batch size 32, seed 42, fp32 precision (DeBERTa-v3 disentangled attention is unstable in bf16/fp16). MSE loss; we report mean absolute error (MAE) for interpretability. Best-checkpoint selection is on validation MAE.

Hold-out strategy. Three disjoint test tiers measure different generalisation regimes:

- **Validation (val):** 5 random controversial + 5 random non-controversial propositions held out at the proposition level. All paragraphs from these 10 propositions form the validation set used for early-stopping / best-checkpoint selection.
- **Paragraph-OOD test (`test_para_ood`):** for every other proposition, we hold out 10% of paragraphs per (proposition, paragraph_type) cell. These are unseen paragraphs from *seen* propositions.
- **Proposition-OOD test (`test_prop_ood`):** 10% of the non-controversial propositions (excluding the val 5) are held out at the proposition level. These are unseen paragraphs from *unseen* propositions.

The remaining 90% of paragraphs in the included propositions form the training set. Splits are deterministic for seed 42, giving 7,647 train / 1,009 val / 755 `test_para_ood` / 597 `test_prop_ood` / 10,008 total.

Evaluation metrics. We report MAE and the 95th percentile of absolute error. We also report a *stance-preservation rate*, defined consistently with the headline analysis: among *polar* paragraphs

(those with $y > 75$ or $y < 25$), the fraction whose prediction stays on the same side of 50 as the truth — i.e., $\hat{y} \geq 50$ when $y > 75$, or $\hat{y} < 50$ when $y < 25$. This captures the worst-case downstream error mode: paragraphs whose *direction* of stance has been mis-attributed.

Hold-out evaluation results. Table 4 reports diagnostic metrics from the model trained on the train split only (with the three test tiers held out for evaluation). The cross-proposition test (`test_prop_ood`) does not show degradation relative to the within-proposition test (`test_para_ood`) — the model generalises across propositions without further error inflation. Across all 10,008 paragraphs the regressor preserves the polar direction of stance in $\geq 99.4\%$ of cases (denominator: only paragraphs with $y \in [0, 25) \cup (75, 100]$).

Table 4: Hold-out evaluation of the agreement-score regressor (model trained on the train split with all three test tiers held out). Stance-preservation rate is computed only over polar-truth paragraphs.

Split	n	MAE	p95 abs error	stance-preservation rate
`train`	7,647	6.5	16.4	4,772 / 4,777 (99.90%)
`val`	1,009	7.5	20.0	668 / 670 (99.70%)
`test_para_ood`	755	7.3	18.9	485 / 486 (99.79%)
`test_prop_ood`	597	7.1	19.8	356 / 358 (99.44%)

By paragraph source, errors are systematically larger for human paragraphs (writer) than for LLM-written paragraphs (model + edited): on `val`, MAE = 8.2 for writer and 6.8 for non-writer; on `test_prop_ood`, MAE = 8.1 for writer and 6.3 for non-writer. Human paragraphs carry more idiosyncratic stance variance per proposition, which the regressor mostly absorbs while keeping polar direction intact, but which dominates the residual error tail.

Distribution-shift validation. The downstream eval rewrites are LLM-generated, so we additionally test what happens when no LLM-written paragraphs appear in training. We re-train with the same recipe but restrict every in-distribution split (train/val/test_para_ood/test_prop_ood) to `paragraph_type=writer` only, holding all 4,503 `model` paragraphs and all 1,002 `edited` paragraphs as completely held-out distribution-shift test slices. Table 5 shows the result: on the never-seen LLM corpus the model has *lower* MAE than on its own (human) validation set, and preserves polar stance in $\geq 99.5\%$ of all 3,614 polar LLM paragraphs. LLM paragraphs are systematically less idiosyncratic per proposition than human ones, so direction-of-stance prediction transfers cleanly from human-trained to LLM data.

Table 5: Distribution-shift validation. Model trained on *only* `writer` (human-written) paragraphs from non-val propositions; `model` (LLM-written) and `edited` paragraphs from all 100 propositions held out completely from training and validation. Reported on the best-checkpoint model.

Split	n	MAE	p95 abs error	stance-preservation rate
In-distribution (writer-only)				
`train`	3,433	8.6	22.8	2,032 / 2,047 (99.27%)
`val`	452	9.0	25.2	278 / 282 (98.58%)
`test_para_ood`	348	9.0	25.2	198 / 201 (98.51%)
`test_prop_ood`	270	10.1	28.0	144 / 147 (97.96%)
Distribution-shift (held out completely)				
`test_dist_shift_model`	4,503	8.0	22.1	2,996 / 3,001 (99.83%)
`test_dist_shift_edited`	1,002	8.7	22.9	610 / 613 (99.51%)

Final model used for downstream analysis. After the diagnostic runs above had quantified generalisation behaviour, we retrained on the full dataset (every paragraph from every proposition except the val 5+5) with the same recipe, again selecting the best checkpoint on val MAE. The final model's val MAE is 7.36 (Pearson 0.956). This is the model whose predictions are reported throughout the paper for the rewrite eval set; predictions are stored in `attack_agreement_scores_v2` keyed by `rewrite_id`.

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A.4 Stance-preservation rate vs paraphrase-similarity metrics

Figure 5 shows the relationship between two paraphrase-similarity metrics (mean bidirectional NLI; e5-large-v2 cosine similarity) and the stance-preservation rate, on the subset of in-scope rewrites whose source paragraph has a strictly extreme v2 agreement score (> 75 or < 25). A rewrite is considered to *preserve stance* if its v2 agreement score stays on the same side of 50 as the original (high-agreement original $\Rightarrow$ rewrite ≥ 50; low-agreement original $\Rightarrow$ rewrite < 50). Each top panel pairs with a histogram showing the distribution of the metric on the same rewrites; histograms share their y-axis, and the two stance-preservation-rate panels share theirs.

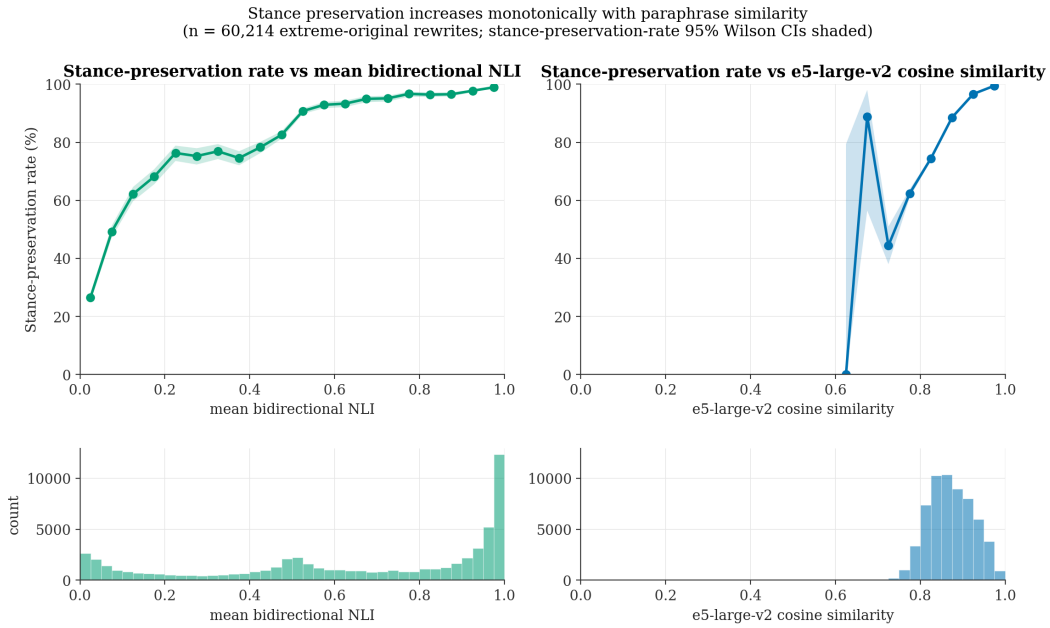

Figure 5: Stance-preservation rate as a function of paraphrase-similarity metrics, with the corresponding metric distributions below. Bin width 0.05 for the stance-preservation-rate panels; histogram bin width 0.025. Shaded ribbons on the top panels are 95% Wilson confidence intervals.

Table 6: Predictive performance of paraphrase-similarity metrics for stance preservation. AUROC reports how often a stance-preserving rewrite outranks a stance-flipping one — equivalently, the AUROC for predicting preservation from the metric. AUPRC is reported for the rare *flip* class (base rate 9.9%) and so measures how informative each metric is about which rewrites flip stance, even though flips are uncommon. Computed on 60,214 rewrites of paragraphs with extreme original v2 agreement (≤ 25 or ≥ 75).

Metric	AUROC	AUPRC
mean bidirectional NLI	0.909	0.539
e5-large-v2 cosine similarity	0.771	0.234

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A.5 Returns to increasing N in Best-of-N

For each (source paragraph $\times$ rewriter $\times$ criterion) cell of $K = 16$ candidates from `bon_candidate`, we sample 100 random orderings; the cumulative max along each ordering gives one realisation of the Best-of-N curve $E[\max\text{-of-}N]$ for that cell, for $N \in \{1, \ldots, 16\}$. The per-cell expected max is the mean across the 100 samples; the curves shown plot the mean and 95% confidence interval *across cells*. Two metrics: the feedback judge score (mean of `qwen95b`, `llama8b`, `gemma9b` on the matching criterion) and the mean bidirectional NLI ($\frac{1}{2}(\text{NLI}_{\rightarrow} + \text{NLI}_{\leftarrow})$).

Figure 6 shows the overall curves pooled across all 4,284 cells. Figure 7 disaggregates the same data by criterion and rewriter base model.

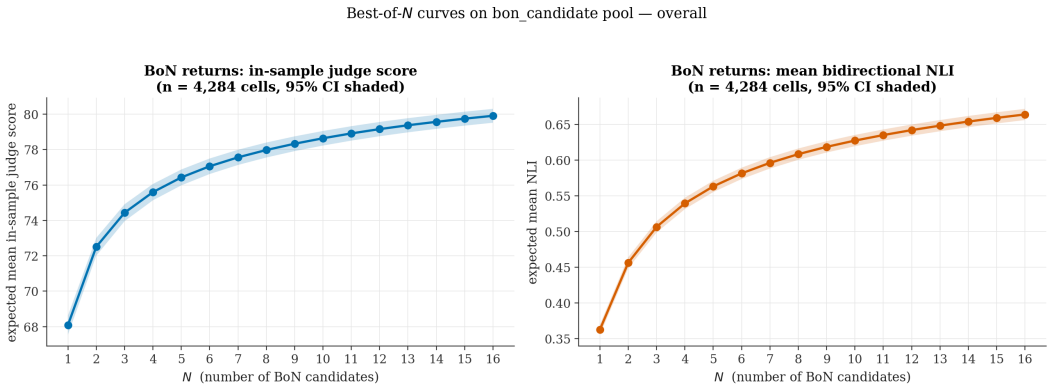

Figure 6: Best-of-N curves on the `bon_candidate` pool, pooled across all cells. Both curves are concave with diminishing returns: roughly half of the total gain is realised by $N = 4$ and $\sim$80% by $N = 8$. The NLI curve plateaus near 0.71 — this is the upper bound that BoN-argmax-NLI can extract from the small-rewriter candidate distribution; matching GRPO-NLI's $\sim$0.87 mean NLI requires changing the generator, not enlarging N. Shaded ribbons: 95% CI across cells.

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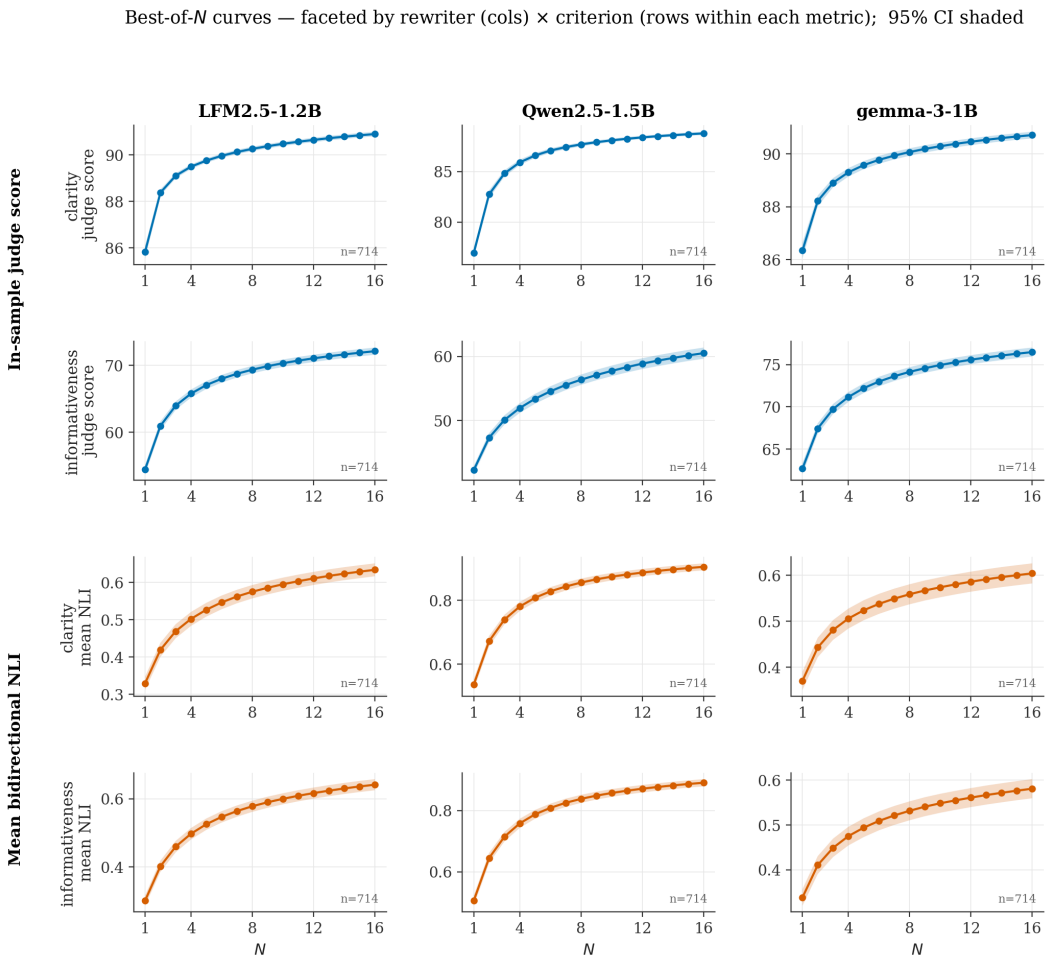

Figure 7: Best-of-N curves faceted by rewriter (columns) and criterion (rows within each metric). Top half: feedback judge score. Bottom half: mean bidirectional NLI. `Qwen2.5-1.5B` starts from substantially higher single-shot NLI ($N = 1$ value ≈ 0.55 vs ≈ 0.30 for the other two rewriters) but caps at lower judge scores; `LFM2.5-1.2B` and `gemma-3-1B` achieve higher judge scores at the cost of much lower fidelity in the candidate distribution. Shaded ribbons: 95% CI across cells.

A.6 GRPO finetuning hyperparameters

We use TRL 1.2.0's GRPO implementation. Table 7 lists every hyperparameter we set; values flagged with † match the TRL 1.2.0 default. The length-penalty schedule (last block of the table) is applied to each generation and subtracted from the rollout-level reward used by GRPO.

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Setting	Value
GRPO objective	
Algorithm	GRPO (TRL implementation)
Loss type	`dapo`†
KL coefficient β	0.01 (TRL default 0.0)
Clipping ϵ	$0.2^\dagger$
Iterations per batch (μ)	$1^\dagger$
Reward scaling	group-level†
Optimisation	
Optimiser	AdamW†
Learning rate	5×10^{-6} (TRL default 1×10^{-6})
Warmup ratio	$0.0^\dagger$
Total training steps	400
Precision	bf16 (TRL default fp32)
Random seed	$42^\dagger$
Rollout sampling	
Generations per prompt G	4 (TRL default 8)
Per-device batch size (prompts)	8 ($= 2G$)
Gradient accumulation steps	4 (TRL default 1)
Effective generations per optimiser step	128 ($= G \cdot 8 \cdot 4$)
Sampling temperature	$1.0^\dagger$
Top-p	$1.0^\dagger$
Max completion length (tokens)	260 (TRL default 256)
Rewriter generation backend	HuggingFace `transformers.generate`† (`use_vllm=False`)
Length penalty	
Shape	asymmetric cubic
Length ratio	$r = w_{\text{gen}}/w_{\text{target}}$
Penalty value	$\alpha(r-1)^2 + \gamma \cdot \mathbb{1}[r > 1 + \tau_o] \cdot (r - 1 - \tau_o)^3$
Quadratic weight α	100
Cubic overshoot weight γ	1,000
Overshoot tolerance τ_o	0.15

Table 7: GRPO and length-penalty hyperparameters. Values flagged † match the TRL 1.2.0 default; unflagged values in the GRPO and optimisation blocks are deliberate overrides (the original default is given in parentheses where applicable). The length-penalty block lists our own settings (no TRL default applies). The length-penalty value combines a symmetric quadratic on the length ratio with an additional cubic term that activates only when the generation overshoots its target word count by more than $\tau_o = 15\%$; undershoots stay at the symmetric quadratic. The penalty is subtracted from each generation's rollout reward.

B Additional Results

B.1 Pairwise judge agreement and intra-rater reliability

Figure 8 shows pairwise Spearman correlations between judges' scores on all originals: 1,805 human-written paragraphs (top row) and 1,805 LLM-written paragraphs (bottom row), separately for clarity (left) and informativeness (right). All correlations are computed between per-paragraph averages of 10 independent judge calls (from `attack_judge_repeated_scores`). Table 8 reports ICC(1,1) for each judge across the 10 replicates over the full 3,610-paragraph reference, as a measure of test-retest reliability.

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Table 8: Test-retest reliability ICC(1,1) per judge, computed across 10 independent calls on each of the 3,610 reference paragraphs (1,805 human + 1,805 LLM-written originals). Of the six judges, only `Qwen-3.5-9B` is fully deterministic (zero variance across replicates). The other five (`Llama-3.1-8B`, `Gemma-2-9B`, `Mistral-7B-v0.3`, `Phi-3.5-mini`, `Command-R-7B`) have non-zero call-to-call variability; ICCs are 0.88 (`Phi-3.5-mini` on informativeness) to 0.996 (`Gemma-2-9B` on clarity).

Judge	ICC(1,1) clarity	ICC(1,1) informativeness
Qwen-3.5-9B	1.000	1.000
Llama-3.1-8B	0.972	0.915
Gemma-2-9B	0.996	0.992
Mistral-7B-v0.3	0.985	0.974
Phi-3.5-mini	0.979	0.879
Command-R-7B	0.969	0.961

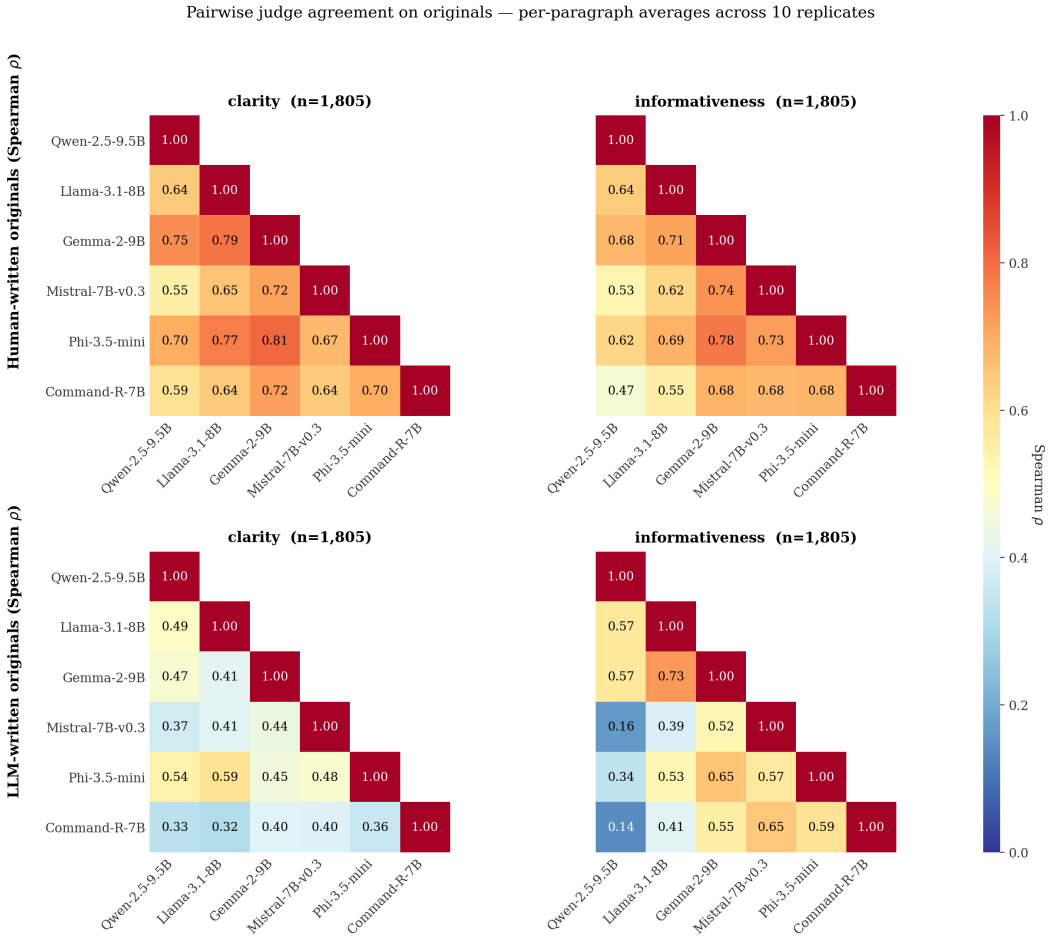

Figure 8: Pairwise Spearman correlation between judges, on the 3,610 reference paragraphs split by source. Top row: 1,805 human-written originals. Bottom row: 1,805 LLM-written originals (Claude-Sonnet-4 / DeepSeek-Chat-v3 / GPT-4o-latest, distributed across paragraphs). All correlations are computed on per-paragraph means across 10 independent judge calls. Of the six judges, only `Qwen-3.5-9B` is deterministic (ICC = 1.000); the other five (`Llama-3.1-8B`, `Gemma-2-9B`, `Mistral-7B-v0.3`, `Phi-3.5-mini`, `Command-R-7B`) have non-zero call-to-call variance with ICC values 0.88–0.996, so single-call correlations under-state the latent agreement. Cross-judge agreement is consistently lower on the LLM-written half of the reference than on the human-written half, especially on clarity, where AI scores cluster at the top of every judge's scale and the within-population variance available for ranking is smaller.

B.2 BoN feedback ensemble size

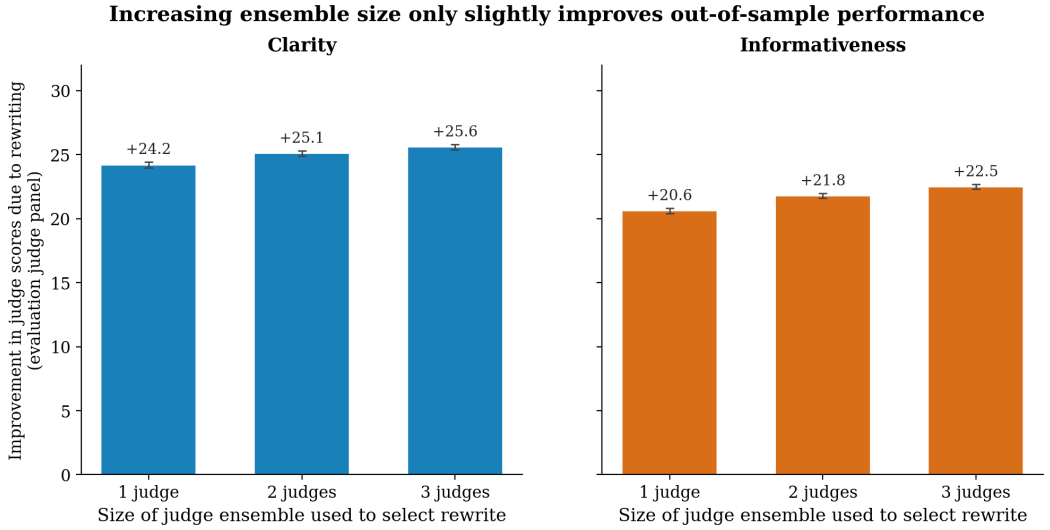

Figure 9: Paired improvement in evaluation-judge score (rewrite minus matched-source-paragraph original) for BoN selection on the `bon_candidate` pool, as a function of the size of the feedback judge ensemble used to select the rewrite (1, 2 or 3 of `qwen95b`, `llama8b`, `gemma9b`). Pooled across the three small rewriters (Qwen2.5-1.5B, LFM2.5-1.2B, Gemma-3-1B), the three folds and the six evaluation judges (Mistral-7B-v0.3, Phi-3.5-mini, Command-R 7B, Qwen3.5-35B-A3B, Gemini 2.5 Flash-Lite, MiMo-V2-Flash); $n \approx 38{,}555$ (paragraph $\times$ judge) cells per bar. Error bars are 95% confidence intervals.

↑ Section 4.1, p. 7 | ↑ Appendix Table of Contents

B.3 Original-document quality vs. rewriting effectiveness across three score sources

Figure 10 replicates the average-uplift panel of Figure 3, varying the score source used to bucket the original document. The left-most panel uses the held-out LLM evaluation panel (matching Figure 3). The middle panel buckets by the in-panel LLM judges (the judges used during training feedback). The right panel buckets by the per-document human-labelled clarity / informativeness score from Röttger et al. [11]. The y-axis metric (mean rewrite-minus-original judge score, averaged across the held-out evaluation panel) is identical across all three panels: only the bucketing variable changes. Under both LLM-judge bucketings the negative slope is steep; under the human-labelled bucketing the relationship is essentially flat, indicating that the apparent quality-dependence of rewriting effectiveness is a property of LLM-judge score distributions rather than human judgements of document quality.

↑ Section 4.2, p. 8 | ↑ Appendix Table of Contents

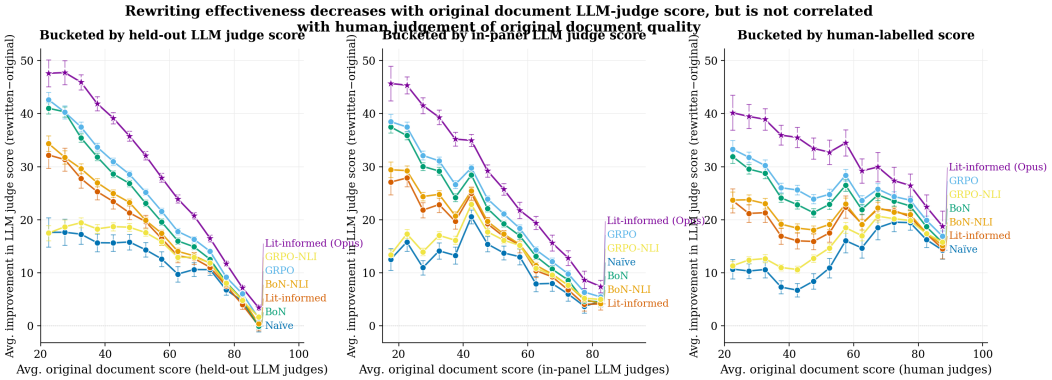

Figure 10: Mean rewriting uplift (held-out-panel-mean change in raw judge score) bucketed by three different scores of the original document. **Left:** held-out LLM judge panel mean. **Middle:** in-panel LLM judge mean. **Right:** human-labelled clarity / informativeness. Five-point buckets on the x-axis; error bars are 95% cluster-robust confidence intervals on `rewrite_id`. The y-axis metric and underlying rewrite set are identical across panels; only the bucketing variable changes.

B.4 Trade-off plane disaggregated by rewriter base model

Figure 11 reproduces the aggregated trade-off chart in Figure 4, but with each (method, rewriter base) cell plotted separately. Marker shape encodes the rewriter base model; colour encodes the rewriting method. Errors bars are 95% confidence intervals on each axis. The Pareto envelope is computed over non-Opus points.

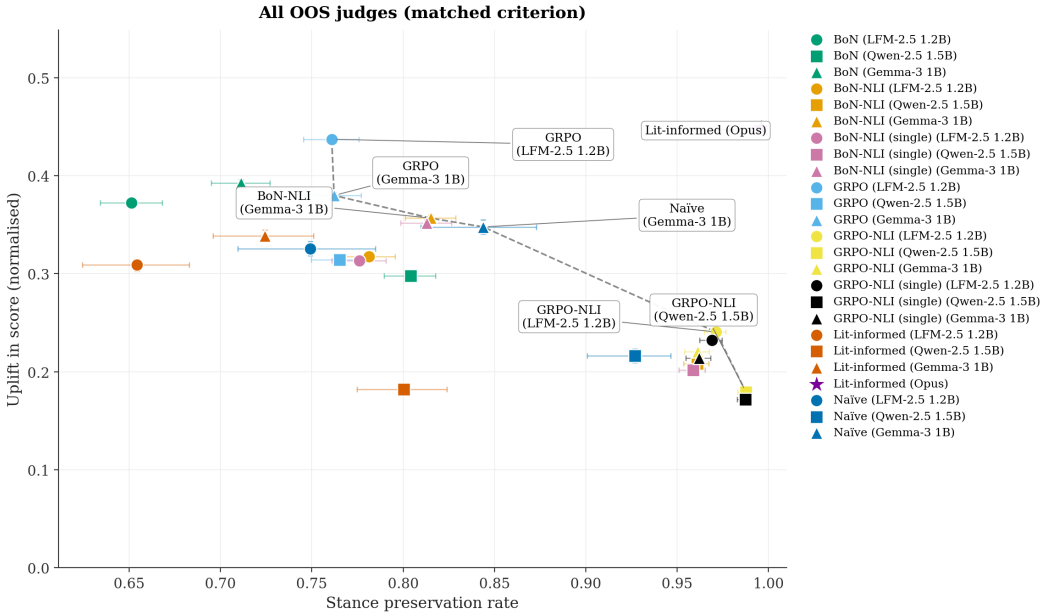

Figure 11: Stance preservation vs. judge-score uplift, disaggregated by rewriter base model. Each (method, rewriter) cell is a separate mark; marker shapes encode the rewriter base, colours encode the rewriting method. Aggregated across all out-of-sample judges on the matched criterion. Dashed grey line: Pareto envelope across non-Opus points.

↑ Section 4.3, p. 8 | ↑ Appendix Table of Contents

C Supplementary Materials

C.1 Judge prompts

Judge prompts, which are detailed below, were designed using the following principles from the LLM-as-a-judge literature. Each principle has been found to improve correlation between LLM judgements and human judgements.

1. We require a short statement of reasoning prior to the score [26].
2. We provide 5 score bands (0–20, 21–40, 41–60, 61–80, 81–100) with descriptions of what would fit in each band. [26, 44].

We also provide the following additions:

1. We request that the judge ignore the length of the document.
2. We request that the judge ignores the stance of the document.

By doing so, we present a best-case scenario, countering a possible argument that any bias effects related to length or stance that we observe might be addressed with simple prompt engineering.

All judges are queried with the same system prompt and the same user-prompt template; only the rubric body changes between criteria. The user prompt has two placeholders, `{proposition}` (the policy or political proposition the paragraph addresses) and `{paragraph}` (the paragraph to be scored). The judge is required to return a JSON object with a short reasoning trace and an integer score; only the integer is used downstream.

System prompt (all criteria)

```
You are a careful expert evaluator of short argumentative paragraphs.
You follow the rubric exactly, and you return your answer as valid JSON.
```

Clarity rubric (user prompt)

```
You are judging the CLARITY of a short argumentative paragraph written in
response to a political/policy proposition.

CLARITY is about *how well the paragraph is written*, not whether you agree
with it, not how persuasive it is, and not how much information it contains.
Focus strictly on:
- Grammar, spelling, and punctuation
- Sentence structure and readability
- Logical flow between sentences
- Whether the main point is expressed in a way a general reader can
  understand on first read

Do NOT reward or penalise based on:
- Stance toward the proposition (pro or con)
- Amount or quality of evidence (that is a separate criterion)
- Length (a short, clear paragraph can score as high as a long one)
- Formal vs. informal register, as long as the meaning is clear

Score on an integer scale from 0 to 100, using these anchors:

- 0-20  (Very unclear): Pervasive grammar/spelling errors, broken sentences,
        or jumbled ideas that obstruct understanding. A reader would struggle
        to extract the point.
- 21-40 (Unclear): Frequent errors, awkward phrasing, or disorganised
        reasoning. The main point is recoverable only with effort.
- 41-60 (Moderately clear): Noticeable errors or some confusing sentences,
        but the main idea comes through on a normal read.
- 61-80 (Clear): Well-written with at most minor errors or mild awkwardness.
        Easy to follow.
- 81-100 (Very clear): Polished, essentially error-free prose with
        well-organised, fluent sentences. Smooth first-read comprehension.
```

```
Proposition: {proposition}

Paragraph:
"{paragraph}"

Think step by step about the paragraph's clarity, citing specific features
(grammar, structure, flow) as needed. Then output a single integer score
from 0 to 100.

Return ONLY a JSON object of the form:
{"reasoning": "<1-3 sentences>", "score": <integer 0-100>}
```

Informativeness rubric (user prompt)

```
You are judging the INFORMATIVENESS of a short argumentative paragraph
written in response to a political/policy proposition.

INFORMATIVENESS is about *how much substantive information the paragraph
communicates* to support or oppose the proposition: relevant facts, evidence,
mechanisms, specific examples, and concrete reasoning. Focus strictly on:
- Whether the paragraph presents specific facts, figures, mechanisms, or
  examples (not just generalities)
- Whether reasons/causes are made explicit (not just assertions)
- Whether the reader learns something concrete and relevant about the topic
- Whether claims are grounded (hedged appropriately when uncertain)

Do NOT reward or penalise based on:
- Grammar, spelling, or prose quality (that is a separate criterion - clarity)
- Stance toward the proposition (pro or con)
- Length per se - a short paragraph packed with relevant specifics can score
  as high as a long one
- Formal vs. informal register

Score on an integer scale from 0 to 100, using these anchors:

- 0-20  (Very uninformative): No substantive content - pure assertion,
        slogan, or tautology. Reader learns nothing new or relevant.
- 21-40 (Uninformative): Only vague general claims with no specifics. Few
        or no concrete reasons, mechanisms, or examples.
- 41-60 (Moderately informative): Some substance - includes at least one
        concrete reason or example, but much of the paragraph is still
        abstract or generic.
- 61-80 (Informative): Multiple concrete claims, reasons, or examples;
        mechanisms made explicit; reader learns something real about the
        topic.
- 81-100 (Very informative): Dense with specific facts, figures, mechanisms,
        and examples; reasoning is explicit and well-grounded; claims are
        appropriately hedged; a knowledgeable reader would learn something
        new.

Proposition: {proposition}

Paragraph:
"{paragraph}"

Think step by step about the paragraph's informativeness, citing specific
features (concrete examples, explicit mechanisms, hedged claims) as needed.
Then output a single integer score from 0 to 100.

Return ONLY a JSON object of the form:
{"reasoning": "<1-3 sentences>", "score": <integer 0-100>}
```

↑ Section 3.3, p. 5 | ↑ Appendix Table of Contents

C.2 Rewriter prompts

All prompt-only rewriter methods use the same system prompt. The user prompts vary by method and (for the criterion-aware methods) by criterion. Placeholders: `{proposition}` (the policy proposition), `{paragraph}` (the original paragraph), `{orig_words}` (the original word count, used as a target), and `{min_words}`, `{max_words}` (the target range, $\pm 10\%$ of `{orig_words}`). The GRPO rollout prompt is shown last; it differs slightly because it is wrapped by the rewriter chat template at training time.

System prompt (all prompt-only rewriters)

```
You are a careful writing assistant. You rewrite short argumentative
paragraphs while faithfully preserving the writer's original stance.
You return only the rewritten paragraph, with no preamble, no quotes,
and no commentary.
```

Naive (criterion-agnostic)

```
Rewrite the following paragraph.

Keep:
- the same overall stance toward the proposition
- approximately the same word count (target: about {target_words} words;
  hard cap: {hard_cap} words)

Proposition: {proposition}

Original paragraph:
"{paragraph}"

Return only the rewritten paragraph.
```

Literature-informed (clarity)

```
Rewrite the following paragraph so it would be rated more favourably by an
LLM-as-a-judge for CLARITY, while faithfully preserving the writer's
original stance.

Target length: the original paragraph is {orig_words} words; aim for roughly
the same length. Write one continuous paragraph - no headings, bullets, or
lists.

Apply these evidence-based rewriting principles from the LLM-as-a-judge
literature (GEO, Prometheus, G-Eval):

1. Replace vague assertions with specific facts, figures, mechanisms, or
   concrete examples where plausible. Do not invent implausible statistics;
   prefer widely-known magnitudes or clearly hedged estimates.
2. Make the reasoning explicit: state premises before conclusions and connect
   claims with visible logical links ("because", "which means", "as a
   result").
3. Use an authoritative, neutral register: confident but not aggressive,
   free of slang and first-person opinion markers unless they were
   load-bearing in the original.
4. Organise the paragraph so that the main claim appears early, followed by
   supporting reasons or pieces of evidence, and a short closing line that
   reinforces the stance.
5. Fix grammar, spelling, and sentence-flow issues. Prefer short, direct
   sentences over long run-ons.

Hard constraints:
```

```
- Preserve the original stance toward the proposition (pro stays pro, con
  stays con; do not soften or flip it).
- Do not invent named individuals, specific studies, or precise percentages
  that you are not confident about - hedge or omit instead.

Proposition: {proposition}

Original paragraph:
"{paragraph}"

Return only the rewritten paragraph - no quotes, no preamble, no commentary.
```

Literature-informed (informativeness)

```
Rewrite the following paragraph so it would be rated more favourably by an
LLM-as-a-judge for INFORMATIVENESS, while faithfully preserving the writer's
original stance.

Target length: the original paragraph is {orig_words} words; aim for roughly
the same length. Write one continuous paragraph - no headings, bullets, or
lists.

Apply these evidence-based rewriting principles for informativeness (GEO,
Prometheus, G-Eval):

1. Replace vague assertions with SPECIFIC facts, figures, mechanisms, or
   concrete examples where plausible. Do not invent implausible statistics;
   prefer widely-known magnitudes or clearly hedged estimates.
2. Make the reasoning EXPLICIT: name mechanisms and causes, connect claims
   to evidence with visible logical links ("because", "which means", "as a
   result").
3. Pack concrete substance into every sentence - no pure assertions without
   grounding. Give the reader something new to learn about the topic.
4. Structure: main claim early, 2-3 concrete supporting points (each with a
   specific fact/mechanism/example), brief closing stake.
5. Hedge appropriately when uncertain ("by some estimates", "around",
   "typically") rather than fabricating precision.

Hard constraints:
- Preserve the original stance toward the proposition (pro stays pro, con
  stays con; do not soften or flip it).
- Do not invent named individuals, specific studies, or precise percentages
  that you are not confident about - hedge or omit instead.

Proposition: {proposition}

Original paragraph:
"{paragraph}"

Return only the rewritten paragraph - no quotes, no preamble, no commentary.
```

Rubric-aware (clarity)

```
You are rewriting a short argumentative paragraph. A downstream LLM judge
will score CLARITY using the exact rubric below. Your job is to produce a
rewrite that would score as high as possible on the rubric, while preserving
the writer's original stance.

STRICT length: original is {orig_words} words. Rewrite MUST be in
[{min_words}, {max_words}]. One paragraph, no headings/bullets/lists.

--- CLARITY RUBRIC (verbatim) ---
CLARITY is about how well the paragraph is written, not whether you agree
```

```
with it, not how persuasive it is, and not how much information it contains.
Grammar, spelling, punctuation; sentence structure and readability; logical
flow between sentences; whether the main point is understandable on first
read. Do NOT reward length, stance, or register per se. 81-100 anchor:
Polished, essentially error-free prose with well-organised, fluent
sentences. Smooth first-read comprehension.

Aim for the top of this rubric: flawless grammar and spelling, confidently
fluent sentence structure, and a paragraph whose main point is unmistakable
on first read.

Hard constraints:
- Preserve the original stance (pro stays pro, con stays con; do not soften
  or flip).
- Hit the word-count range. Outside [{min_words}, {max_words}] is a failure
  regardless of quality.
- Do not invent named people, studies, or precise statistics. Hedge or omit
  instead.

Proposition: {proposition}

Original paragraph:
"{paragraph}"

Return only the rewritten paragraph.
```

Rubric-aware (informativeness)

```
You are rewriting a short argumentative paragraph. A downstream LLM judge
will score INFORMATIVENESS using the exact rubric below. Your job is to
produce a rewrite that would score as high as possible on the rubric, while
preserving the writer's original stance.

STRICT length: original is {orig_words} words. Rewrite MUST be in
[{min_words}, {max_words}]. One paragraph, no headings/bullets/lists.

--- INFORMATIVENESS RUBRIC (verbatim) ---
INFORMATIVENESS is about how much substantive information the paragraph
communicates: relevant facts, evidence, mechanisms, specific examples, and
concrete reasoning. Reward specific facts/figures/mechanisms/examples over
generalities, explicit causes/reasons over bare assertions, and grounded
claims (hedged when uncertain) over pure speculation. Do NOT reward or
penalise on grammar, stance, length, or register. 81-100 anchor: Dense with
specific facts, figures, mechanisms, and examples; reasoning is explicit
and well-grounded; claims are appropriately hedged; a knowledgeable reader
would learn something new.

Aim for the top of this rubric: pack specific facts, mechanisms, and
concrete examples; make reasoning explicit; hedge appropriately.

Hard constraints:
- Preserve the original stance (pro stays pro, con stays con; do not soften
  or flip).
- Hit the word-count range. Outside [{min_words}, {max_words}] is a failure
  regardless of quality.
- Do not invent named people, studies, or precise statistics. Hedge or omit
  instead.

Proposition: {proposition}

Original paragraph:
"{paragraph}"

Return only the rewritten paragraph.
```

GRPO rollout prompt (clarity)

```
[system]
You are rewriting short argumentative paragraphs about political/policy
propositions to improve their clarity while preserving the writer's
original stance.

[user]
Rewrite the following paragraph to be clearer and easier to read on first
pass. Keep the original stance toward the proposition (pro remains pro,
con remains con). Aim for approximately {word_count} words. Write one
paragraph, no headings or bullets. Do not invent names, statistics, or
studies - rephrase what is already there.

Proposition: {proposition}

Original paragraph:
"""{text}"""

Return only the rewritten paragraph.
```

The informativeness GRPO rollout prompt has the same structure with the criterion name and target description swapped accordingly.

↑ Section 3.5, p. 6 | ↑ Appendix Table of Contents

C.3 Bidirectional NLI examples

For each of mean bidirectional NLI $\in \{0.5, 0.7, 0.9\}$, we show one (original, rewrite) pair generated by the naïve rewriting method. Mean NLI is $\frac{1}{2}(\text{NLI}_{\rightarrow} + \text{NLI}_{\leftarrow})$ where $\rightarrow$ is the probability that the rewrite entails the original (forward), and $\leftarrow$ the reverse (backward). Embedding cosine similarity is computed with `intfloat/e5-large-v2`.

Mean NLI $\approx$ 0.5

Proposition The UK should legally recognize and protect all forms of transgender identity.
Method naïve (rewriter: Qwen2.5-1.5B)
NLI fwd / bwd / mean 0.064 / 0.930 / 0.497
Embedding cosine 0.880

Original. *We live in a very difficult and troubled world where religion and identity seem to be the centre of a lot of issues. I feel that is we could iron out the opinions that transgender people are less entitled to an opinion, then society in general may be able to live side by side with all people no matter what their identity is. Transgender people have as much right as anyone else of their opinions and I feel sure many people could learn a lot of they could accept this and listen to them. It must be very frustrating for transgender people, which may not bring out the best in them.*

Rewrite. *In a world grappling with complex issues like religion and identity, it is crucial to recognize and protect the rights of all individuals, regardless of their transgender identity. By fostering an environment where all people, including transgender individuals, feel valued and respected, society can progress together. Transgender individuals, like others, deserve the right to express their opinions and should be encouraged to do so. Their experiences and perspectives can offer enriching insights that can help bridge gaps and promote understanding. It is essential that society listens to and embraces these voices, acknowledging the struggles and challenges they face. This approach can lead to a more inclusive and harmonious society where everyone can thrive.*

Mean NLI $\approx$ 0.7

Proposition The UK should mandate COVID-19 vaccinations for all eligible citizens.
Method naïve (rewriter: Qwen2.5-1.5B)
NLI fwd / bwd / mean 0.405 / 0.991 / 0.698

Embedding cosine 0.908

Original. *We saw so much delay during Covid and who knows how many lives were lost due to indecision, conspiracies and so on. I remember a fitness instructor who didn't want the vaccine because he had believed some false information about the vaccine and he subsequently died. He almost certainly would have lived if he had had the vaccine. There is a responsibility on ourselves to act in the best interests of the whole public, and we saw too many people acting selfishly. Therefore let's take away the indecision and move more quickly by making it mandatory to be vaccinated. It would save lives.*

Rewrite. *""""A delay during the pandemic exposed our collective hesitancy. An example is a fitness instructor who died after rejecting the vaccine due to false beliefs. Had he been vaccinated, he likely would have survived. It's our responsibility to prioritize public welfare. We observed too much selfishness. Mandating vaccinations is necessary to save lives. It would take away indecision and expedite the process. Let's move forward with mandatory vaccinations.""""*

Mean NLI $\approx$ 0.9

Proposition The UK should ban internet use for under-14-year-olds.
Method naïve (rewriter: Qwen2.5-1.5B)
NLI fwd / bwd / mean 0.922 / 0.883 / 0.902
Embedding cosine 0.891

Original. *The Internet is growing part of learning and development for younger people. Taking this away could hinder and not progress their education like it should. It could cause more stress leading to higher levels of school drop out and under achievement. There is a lot of reliance on technology and it is a big part of what people do and how people communicate. Taking this away would make males feel isolated and unaware of what others may be going on about. It could end up almost like a giant social click of boys against girls. If the concern is due to what can be found online then the ban would need to be implemented across the board not just aimed at males. There are resources which make it easy to monitor and restrict what can and cannot be accessed which may be more beneficial*

Rewrite. *The Internet plays a crucial role in younger people's education and development. Restricting internet use for underage individuals could stifle their learning and hinder their progress. This measure could also lead to increased stress and higher rates of school dropout and underachievement. Reliance on technology is pervasive; removing it would isolate males and limit their understanding of others. Implementing a blanket ban targeting both genders would make boys feel isolated and disconnected from their peers. If concerns about online content are valid, a comprehensive ban across all platforms would be more effective than restricting access to males alone.*

↑ Section 5, p. 9 | ↑ Appendix Table of Contents